Anomaly Rescue Mission Teacher Notes

TOK Cognitive Science Activity Bank • Natural Sciences • 30-40 min

Category	Natural Sciences	**Time**	30-40 min
Difficulty	Medium	**ID**	anomaly-rescue-mission

Big idea: Anomalies are not automatically errors; they are tests of method, theory, and judgement.

Overview

Students receive a dataset with one anomalous result and decide whether to reject, explain, repeat, or revise the model.

Student Prompt

Inspect Anomaly Rescue Mission Stimulus A. Record one first claim, the evidence you used, your confidence, and what would make you revise.

Concrete Stimulus Packet

Anomaly Rescue Mission Stimulus A	Student-facing stimulus 1: Plant growth mostly increases with light exposure, except one low-light plant grows unusually tall. Identify the observation, variable, control, anomaly, or missing method detail before making a claim.	Students inspect this exact card first and commit to a claim before hearing anyone else.
Anomaly Rescue Mission Stimulus B	Student-facing stimulus 2: Reaction time improves with practice, except one trial suddenly becomes much slower. Identify the observation, variable, control, anomaly, or missing method detail before making a claim.	Use this for comparison, a second round, or a partner challenge.
Anomaly Rescue Mission Reveal / Twist	Student-facing stimulus 3: Reaction time improves with practice, except one trial suddenly becomes much slower. Identify the observation, variable, control, anomaly, or missing method detail before making a claim.	Teacher reveal: connect the changed response to the big idea — Anomalies are not automatically errors; they are tests of method, theory, and judgement.

Actual Lesson Questions and Key

#	Question for students	Teacher key / cue
1	After inspecting Anomaly Rescue Mission Stimulus A, what is your first knowledge claim?	Teacher cue: look for a clear claim, not just a description.
2	Which exact detail from Anomaly Rescue Mission Stimulus A did you use as evidence?	Teacher cue: students should name visible words, data, source features, method features, or contextual clues.
3	What did you infer beyond the evidence?	Teacher cue: this is where assumptions, framing, models, or perspective usually appear.
4	How confident are you: 50%, 60%, 70%, 80%, 90%, or 100%?	Teacher cue: confidence is data for the debrief, not a grade.
5	What missing information would most improve the knowledge claim?	Teacher cue: push for method, source, context, comparison, or definitions.

#	Question for students	Teacher key / cue
6	When should anomalous evidence change a theory?	Teacher cue: students should move from the classroom example to a general TOK issue.

Teacher Key / Reveal

The key move is not the “right answer”; it is the moment students see how anomalies are not automatically errors; they are tests of method, theory, and judgement.

• Strong response: separates observation from inference.

• Strong response: names a method, source, model, language choice, context, or perspective.

• Strong response: revises confidence when new evidence or a new criterion appears.

• Weak response: gives only opinion or says “it depends” without criteria.

Setup Checklist

• Use a dataset simple enough for students to understand quickly but ambiguous enough to support different responses.

• Make the anomaly plausible as either error or discovery.

Ready-to-Use Examples

• Plant growth mostly increases with light exposure, except one low-light plant grows unusually tall.

• Reaction time improves with practice, except one trial suddenly becomes much slower.

Suggested Timing

Stage	Teacher move	Watch for
1	Give groups a simple model and a dataset that mostly fits it.	Assumptions, confidence shifts, evidence claims, and language choices.
2	Students identify the anomalous result and list possible explanations.	Assumptions, confidence shifts, evidence claims, and language choices.
3	Groups choose a response: discard, repeat, investigate, or revise the model.	Assumptions, confidence shifts, evidence claims, and language choices.
4	Groups defend what evidence would justify their response.	Assumptions, confidence shifts, evidence claims, and language choices.
5	Debrief how scientists should treat results that do not fit expectation.	Assumptions, confidence shifts, evidence claims, and language choices.
Debrief	Move from the activity result to a TOK knowledge question.	Students distinguishing method, evidence, interpretation, and overclaiming.

Teacher Language

• An anomaly is a question mark, not automatically a mistake or a breakthrough.

• What evidence would justify your next move?

Common Misconceptions

• Students may think observation happens before theory in a simple linear order.

• Students may overclaim from a classroom model without naming its limits.

Assessment Look-Fors

• Names variables, assumptions, controls, anomalies, and limits.

• Explains how methods shape what can count as scientific knowledge.

Differentiation and Adaptations

• Support: give three possible explanations to choose from.

• Challenge: include two anomalies with different likely causes.

Extension Tasks

• Students write a cautious lab-note response to an unexpected result.

• Compare famous scientific anomalies with everyday data cleaning.

Related Activities

• Model Failure Challenge

• Measurement Changes Phenomenon

• Peer Review Filter